

Partialling Out and the Propensity Score

Why $\tilde{D}_i = D_i - p(X_i)$

ECON 712 — Supplementary Notes

1 Setup

Let $D_i \in \{0, 1\}$ be a binary treatment and $X_i \in \{0, 1\}$ a binary covariate. Define the propensity score $p(x) = E[D_i \mid X_i = x]$.

We want to show that partialling out X_i from D_i via OLS yields the residual $\tilde{D}_i = D_i - p(X_i)$.

2 Step 1: OLS Fitted Values in the Saturated Regression

Regress D_i on a constant and X_i :

$$D_i = \alpha + \gamma X_i + \text{residual}.$$

Because X_i is binary this regression is *saturated* — it has one free parameter per cell. The OLS estimates are exactly the cell means:

$$\begin{aligned}\hat{\alpha} &= E[D_i \mid X_i = 0] = p(0), \\ \hat{\gamma} &= E[D_i \mid X_i = 1] - E[D_i \mid X_i = 0] = p(1) - p(0).\end{aligned}$$

The fitted value for observation i is therefore

$$\hat{D}_i = p(0) + (p(1) - p(0))X_i.$$

3 Step 2: The Fitted Value Equals $p(X_i)$

Check each cell:

$$\begin{aligned}X_i = 0 : \quad \hat{D}_i &= p(0) + (p(1) - p(0)) \cdot 0 = p(0) = p(X_i). \\ X_i = 1 : \quad \hat{D}_i &= p(0) + (p(1) - p(0)) \cdot 1 = p(1) = p(X_i).\end{aligned}$$

In both cells $\hat{D}_i = p(X_i)$, so the two-coefficient saturated regression perfectly recovers the propensity score.

4 Step 3: The Residual is $\tilde{D}_i = D_i - p(X_i)$

The OLS residual (the partialled-out treatment) is

$$\tilde{D}_i = D_i - \hat{D}_i = D_i - p(X_i).$$

5 Step 4: $\tilde{D}_i$ is Already Demeaned

We verify $E[\tilde{D}_i] = 0$ using the Law of Iterated Expectations:

$$E[\tilde{D}_i] = E[E[D_i - p(X_i) \mid X_i]] = E[E[D_i \mid X_i] - p(X_i)] = E[p(X_i) - p(X_i)] = 0.$$

Since $E[\tilde{D}_i] = 0$, it follows immediately that

$$\tilde{D}_i - E[\tilde{D}_i] = \tilde{D}_i - 0 = \tilde{D}_i = D_i - p(X_i).$$

The partialled-out variable is its own demeaned version.

6 Summary

Proposition 1. *Let $X_i \in \{0, 1\}$ be binary and $p(x) = E[D_i \mid X_i = x]$ be the propensity score. In the saturated OLS regression of D_i on $(1, X_i)$:*

1. *The constant and slope equal $p(0)$ and $p(1) - p(0)$ respectively.*
2. *The fitted value equals $p(X_i)$ for every observation.*
3. *The residual (partialled-out treatment) is $\tilde{D}_i = D_i - p(X_i)$.*
4. *$E[\tilde{D}_i] = 0$, so $\tilde{D}_i$ is already mean-zero.*

Remark 1. The same argument extends to a discrete X_i with K cells: a fully saturated regression includes a dummy for each cell, the fitted value in each cell is the cell mean $p(x)$, and the residual is again $D_i - p(X_i)$. This is the foundation of the Frisch–Waugh–Lovell result used in Angrist (1998).